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1 function displayInspectionOutput(originalImg, reportData)
2 % colors
3 grey = [0.12 0.12 0.12];
4
5 % Create a figure sized moderately to render well in the Live Editor output
6 fig = figure('Name', 'Inspection Results', 'Color', grey, ...
7     'Units', 'normalized', 'Position', [0.1 0.1 0.7 0.6]);
8
9 % Use a 2-row, 3-column layout to stack images while keeping text on the left
10 t = tiledlayout(2, 3, 'TileSpacing', 'compact', 'Padding', 'compact');
11
12 % =====
13 % LEFT SIDE: TEXT & DECISION DASHBOARD (Spans 2 rows, 1 column)
14 % =====
15 axText = nexttile(1, [2 1]);
16 axis(axText, 'off'); % Hide the standard graph axes
17
18 % PASS/FAIL Giant Text
19 if strcmp(reportData.Decision, 'PASS')
20     decisionColor = [0 0.8 0]; % Bright Green
21 else
22     decisionColor = [0.8 0 0]; % Bright Red
23 end
24
25 text(axText, 0.5, 0.85, reportData.Decision, 'FontSize', 40, ...
26     'FontWeight', 'bold', 'Color', decisionColor, 'HorizontalAlignment', 'center');
27
28 % AI Classification Text
29 displayClass = string(reportData.AIClass);
30 aiClassText = sprintf('AI Classification:\n%s', displayClass);
31
32 text(axText, 0.5, 0.65, aiClassText, 'FontSize', 14, 'FontWeight', 'bold', ...
33     'HorizontalAlignment', 'center');
34
35 % Vision Evidence Box
36 evidenceText = sprintf(['--- Vision Evidence ---\n', ...
37     'Confidence Score: %.2f%%\n', ...
38     'Total Tubes: %d\n', ...
39     'Normal Tubes: %d\n', ...
40     'Shape Anomalies: %d\n', ...
41     'Color Anomalies: %d'], ...
42     reportData.Confidence, ...
43     reportData.metrics.TotalTubes, reportData.metrics.NormalTubes, ...
44     reportData.metrics.ShapeAnomalies, reportData.metrics.ColorAnomalies);
45
46 text(axText, 0.5, 0.25, evidenceText, 'FontSize', 12, 'HorizontalAlignment', 'center', ...
47     'BackgroundColor', grey, 'EdgeColor', 'black', 'Margin', 5);
48
49 % =====
50 % RIGHT SIDE (TOP ROW): ORIGINAL & MASK IMAGES
51 % =====
52 % Image 1: Original
53 ax1 = nexttile(2);
54 imshow(originalImg, 'Parent', ax1);
55 title(ax1, 'Original Image', 'FontSize', 12);
56
57
58 % Image 2: Mask
59 ax2 = nexttile(3);
60 imshow(reportData.images.ImageMask, 'Parent', ax2);
61 title(ax2, 'Cleaned Mask', 'FontSize', 12);

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62
63 % =====
64 % RIGHT SIDE (BOTTOM ROW): ANNOTATED IMAGE
65 % =====
66 % Image 3: Object Detection (Spans the bottom 2 columns)
67 ax3 = nexttile(5, [1 2]);
68 imshow(reportData.images.AnnotatedImage, 'Parent', ax3);
69 title(ax3, 'Object Detection', 'FontSize', 12);
70
71 % Main Title
72 title(t, 'Automated Tube Inspection Dashboard', 'FontSize', 18, 'FontWeight', 'bold');
73 end
```